

QuantVision AI Research Report

Bayesian Regime Detection Engine for Indian Equity Markets

This report has been generated automatically using Hidden Markov Models, Bayesian Updating, Particle Filtering, Foundation Models, Regime Switching VAR, Conformal Prediction, Ensemble Learning and Portfolio Simulation.

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Executive Summary

QuantVision AI is an intelligent quantitative research platform designed to analyse Indian equity markets using modern statistical, probabilistic and machine learning techniques. The system integrates Bayesian Learning, Hidden Markov Models, Particle Filtering, Regime Switching Models, Bayesian Deep Learning, Foundation Models, Conformal Prediction and Ensemble Learning to detect market regimes, estimate uncertainty and generate investment recommendations.

Current Market Regime

Unknown

Market Summary

Metric	Value
Initial Investment	100000
Final Portfolio Value	104484.34984221536
Profit/Loss	4484.349842215364
Total Return (%)	4.484349842215364
CAGR (%)	5.771854778038787
Win Rate (%)	100.0
Profit Factor	inf
Sortino Ratio	0.4015314330389658
Calmar Ratio	0.9488521569826789
Sharpe Ratio	0.6525041320216318
Max Drawdown (%)	-6.082986412122528
Number of Trades	6

AI Engine Components

Module	Status
Hidden Markov Model	✓ Implemented
Bayesian Updating	✓ Implemented
Particle Filter	✓ Implemented
Bayesian Deep Learning	✓ Implemented
Foundation Model	✓ Implemented
Regime Switching VAR	✓ Implemented
Conformal Prediction	✓ Implemented
Ensemble Learning	✓ Implemented
Portfolio Simulator	✓ Implemented

Portfolio Performance

Metric	Value
Final Portfolio Value	104484.34984221536
Sharpe Ratio	0.6525041320216318
Sortino Ratio	0.4015314330389658
Calmar Ratio	0.9488521569826789
Profit Factor	inf
Number of Trades	6

Risk Analysis

Parameter	Value
Market Risk	Unknown
Prediction Confidence	Unknown
Prediction Uncertainty	Unknown

Suggested Asset Allocation

Not Available

Investment Recommendation

Not Available

The recommendation has been generated using ensemble learning by combining Bayesian inference, Hidden Markov Models, Deep Learning forecasts, Particle Filtering, Foundation Model estimates, Regime Switching VAR and

Disclaimer

This report has been generated automatically by **QuantVision AI** for educational, research and demonstration purposes. Market forecasts are based on historical market data, probabilistic modelling and machine learning techniques. Predictions do not guarantee future performance. Investors should perform independent research and consult qualified financial advisors before making investment decisions.

Future Enhancements

- Multi-Asset Portfolio Support
- Real-Time News Sentiment Analysis
- Macroeconomic Indicator Integration
- Sector Rotation Analysis
- Transformer-Based Foundation Models
- Reinforcement Learning Portfolio Allocation
- Cloud-Based Deployment

Conclusion

QuantVision AI integrates Bayesian statistics, Hidden Markov Models, Deep Learning, Particle Filtering, Regime Switching, Ensemble Learning and Portfolio Analytics into a unified quantitative research platform. The generated market regime, confidence score, allocation guidance and investment recommendation provide a structured framework for analysing Indian equity markets.

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